

Institute of Actuaries of India

Subject SA2 – Life Insurance

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

- i) For the unit-linked products in India, the commissions and distribution expenses are largely front-loaded. On the other hand, regulations require the charges during the lock-in period to be spread out in an even manner, such that the difference between minimum and maximum charge during the first five years is not more than 50%. As such, this creates a mismatch between the incidence of expenses incurred and charges. The high level of initial expenses end up being recouped over time instead of through up-front charges.

[1.5]

The regulations also stipulate that the reserves to be held for the unit-linked business is the aggregate of (a) unit reserves, being equal to the fund value; and (b) non-unit reserves, calculated based on a discounted cash flow approach, allowing for margins for adverse deviations (MADs) in the projection assumptions. Although the regulations require only the total reserves at a policy level to be non-negative, the Actuarial Practice Standard 2 (APS2) issued by the Institute of Actuaries of India (IAI) require the non-unit reserve itself to be non-negative at a policy level. Therefore, even if the recoverability of high acquisition expenses from future charges can be reliably demonstrated, the non-unit reserves cannot be negative to reflect this.

[1.5]

The regulations also require the total reserves to be not less than the surrender value payable (after applying any surrender penalties, which themselves are capped at a low level) for a given policy. Given these reasons, the level of total reserves held for a unit-linked policy ends up being at a higher level than traditional products.

[0.5]

The Indian regulatory solvency requirements are factor based – i.e. the required solvency capital is expressed as a certain percentage of reserves and a certain percentage of ‘sum at risk’. Given this, even though the percentages specified for unit-linked business are lower than those for traditional business, once applied to a relatively higher level of reserves, the resulting capital requirements does not turn out to be low.

Thus, the combined impact of these factors leads to a significant new business strain for the unit-linked business. This is what makes these products capital intensive in the Indian context.

[1.5]

[5]

- ii) The implications of the proposed ART solution on various items are set out below.

Solvency Ratio

The reinsurer will advance an initial cash lump-sum to the Company. Such a lump-sum will increase the assets of the Company and therefore the available solvency margin (ASM).

This lump-sum (along with the Risk Charge thereon) is repayable to the reinsurer, by sharing 60% of the future statutory profits, if and when they emerge. Since the repayment is contingent on the emergence of the future statutory profits, the Company is not required to set up a liability or a provision towards the same and therefore the ‘reserves’ will remain unchanged. Thus, the required solvency margin (RSM) will also not change.

As such, upon the execution of the treaty, the Company's solvency ratio (i.e. ASM / RSM) will improve.

[2]

Statutory Reserves

The initial cash lump-sum received is in the form of reinsurance commissions. As the statutory reserves are set as unit reserves plus non-unit reserves (calculated on a prospective discounted cash-flow method), the reinsurance commissions have no bearing on the statutory reserves.

The repayment of the cash lump-sum (along with the Risk Charge thereon) is accounted for as reinsurance premiums. However, such a repayment is contingent upon the emergence of future statutory profits. Therefore, such reinsurance premiums are not required to be reflected in the calculation of non-unit reserves.

Overall, there will be no change to the statutory reserves of the Company.

[2]

Embedded Value ("EV")

The Adjusted Net Worth ("ANW") of the Company will increase by the amount of the initial cash lump-sum received.

The Present Value of Future Profits ("PVFP") will decrease, as this will reflect the repayment of the Treaty Balance (along with the Risk Charges) in the form of reinsurance premiums from the future statutory profits of the Company. As the Company uses the Indian Embedded Value (IEV) approach, the reduction in the PVFP is likely to be higher than the increase in the ANW. This is because the Risk Charges payable to the reinsurer will be based on MIBOR + 3%, which are likely to be higher than the risk-free rates used in the calculation of PVFP.

The Company may need to test the impact on Cost of Residual Non-Hedgeable Risks ("CRNHR") due to the proposed ART solution. Ordinarily, the unit-linked profit emergence may be significantly reduced in stress scenarios e.g. adverse mortality, higher surrenders etc. used in the calculation of CRNHR. However, with this ART solution in place, some of these tail risks may end up being passed through to the reinsurer. For instance, if the impact of the shock lapse scenario is such that the resulting profitability turns out to be negative, then it is possible that the Treaty Balance may not be required to be repaid by the sunset date and therefore end up getting forfeited. This may result in a reduction in the CRNHR.

Overall, it is likely that there may be a reduction in the EV due to the proposed ART solution, but the interplay of the elements above will have to be quantified to ascertain the exact impact on the EV.

[4]

[8]

- iii) The justification of the proposed ART solution to the regulator will cover the following aspects.

Risk transfer

From a regulatory standpoint, the most important aspect is that an ART solution reflects significant transfer of risk from the insurer to the reinsurer and that it is not essentially a surrogate loan. For traditional reinsurance treaties, it is typically obvious that there is a transfer of risk (e.g. mortality or morbidity) from the insurer to the reinsurer.

However, analyzing the risk transfer for ART solutions requires a thorough review of the proposed solution.

Under the proposed ART solution, the reinsurer advances a cash lump-sum – called the Treaty Balance – to the insurer in exchange for a 60% share of the future statutory profits, if and when they emerge. The insurer repays the Treaty Balance from its future profits and pays the Risk Charges at a rate of MIBOR + 3% p.a. The treaty has a sunset date of twenty years. It will automatically terminate at this point and any outstanding Treaty Balance will be forfeited.

If the repayment of the Treaty Balance is likely in most scenarios, then there may be a weak argument that there is indeed a significant transfer of risk. Conversely, if the Company can demonstrate that under certain scenarios, the Treaty Balance may not be fully repaid and may ultimately be forfeited on the sunset date, then this supports the argument that there is indeed a significant transfer of risk.

[2]

The Company may want to perform financial projections under various scenarios and demonstrate that forfeiture of the Treaty Balance is plausible. Certain plausible adverse scenarios are outlined below:

- Surrender charges are fairly low in the initial years and cease after the completion of the five policy years. As such, this makes the unit-linked products quite susceptible to lapse risk. Given this, the Company may consider an adverse lapse scenario to ascertain the impact on the statutory profits and therefore the settlement of the Treaty Balance.
- The Company may also consider alternative investment return scenarios. For instance, an immediate 50% drop in equity values or reductions in future 'best estimate' interest rates etc. Under some of these scenarios, the future fund management charges may be significantly lower than those in the base scenario results. The Company may then ascertain the impact on the statutory profits over the sunset period and therefore the possibility of non-settlement of the full Treaty Balance.
- The Company may consider an increased mortality / pandemic scenario. If there are high margins in the mortality risk charges (which support the profitability of the unit-linked business to a large extent), such a scenario may indicate the risks of erosion in such margins. It may set the mortality assumption to be significantly high (say, best estimate X 200% or 300%) to ascertain the impact on the statutory profits over the sunset period and therefore the possibility of non-settlement of the Treaty Balance.

Given that the profitability of unit-linked business is already low in India (due to the cap on charges), any such plausible adverse scenarios may result in erosion of statutory profitability and therefore the Company may not be required to settle the Treaty Balance before the sunset period. This may help support the significant risk transfer requirement of the regulator.

[4]

[Max 6]

Cash financing

The proposed ART solution is structured so that the Company receives an up-front cash lump sum. Typically, the regulator may be concerned if the arrangement is structured in the form of 'IOUs' or a line of credit, as opposed to an up-front cash payment as IOUs / line of credit may not be seen as a high rated capital given the counterparty risks.

However, up-front transfer of cash into an Indian insurer would address this potential regulatory concern.

Up-front cash financing means that the Company does not have a counterparty exposure to the reinsurer. Therefore, such a structure would be more palatable from the regulatory standpoint.

It is also noted that the proposed ART solution does not require the Company to keep the cash in an escrow account until the Treaty Balance is repaid. Again, this may be viewed favorably from the regulatory standpoint.

[1 mark each for an appropriate point discussed]

[Max 3]

Investments of policyholders' funds

At times, when reinsurers construct ART solutions, they may exercise significant control or influence on the investment or asset-liability management (ALM) strategies of the insurer. The intent would be to safeguard the reinsurers' commercial interests. An example would be - a reinsurer requiring the insurer to obtain its prior approval for any significant changes to the investment or ALM strategies of the insurer in relation to the covered block.

However, under the proposed ART solution, the unit-linked funds will continue to be maintained / invested by the Company even though the reinsurer is entitled to 60% of the statutory profits arising therefrom. There is no change from the investment or the ALM standpoint. Premiums are not remitted to the reinsurer but are allocated / invested as per the instructions given by the policyholders.

The proposed ART solution ensures that the policyholders' interests are not adversely impacted by executing the solution and the investment of policyholders' funds will continue to be performed by the Company, in accordance with applicable regulatory guidelines.

[1 mark each for an appropriate point discussed]

[Max 4]

Benefits of capital efficiency

Capital efficiency yields several other benefits:

- The proposed ART solution will improve the solvency ratio of the Company. Generally, this is a good outcome for the policyholders as this enhances the security of their benefits.
- The Company can continue to invest into the business and grow the business e.g. investing into distribution, investing into technology etc. It does not have to limit the new business volumes for capital conservation. This will lead to greater economies of scale in the medium term, which may eventually benefit the shareholders through lower unit costs and policyholders through lower charges.
- The improved capital efficiency will generally enhance the investment freedom for the Company. The proposed solution relates to unit-linked business and the investment freedom may not be directly beneficial for unit-linked funds. However, the investment freedom may be relevant for other lines of business such as participating products. The improved capital efficiency avoids a

situation wherein the Company may be constrained to follow a more prudent investment strategy for other product lines, which may result in lower long term returns to the policyholders.

- The improved capital efficiency may help the Company offer other types of products (such as protection) that may also be capital intensive.

[1 mark for each appropriate point]

[Max 3]

Overall, the proposed ART solution can be justified to the regulator on the following grounds:

- There is a significant transfer of risk, as demonstrated by forfeiture of the Treaty Balance under adverse scenarios;
- The reinsurer advances an up-front cash lump-sum (i.e. no strings attached) to the Company;
- Policyholders' investments are not impacted by the proposed ART solution. Investments and ALM are continued to be managed by the insurer and are in line with the applicable investment regulations; and
- It improves the capital position of the Company, which may benefit the Company (and therefore the policyholders) in the medium term.

[Up to 1 mark to summarise the main points]

[Max 15]

- iv) In accordance with the IRDAI (Expenses of Management of Insurers transacting life insurance business) Regulations, 2016, the following minimum items are required to be covered by the insurers in the policy for allocation of direct expenses and apportionment of indirect expenses:

- expenses which shall be allocated;
- basis of such allocation;
- expenses which shall be apportioned;
- basis of such apportionment;
- manner of allocation and/or apportionment of acquisition expenses and renewal expenses;
- manner in which compliance with the policy for allocation/apportionment of expenses will be ensured

[0.5 marks for each of the items]

[Max 3]

- v) Under the proposed ART solution, the reinsurer is entitled to receive a 60% share of the future statutory profits of the covered unit-linked business.

It is possible that the Company may reflect a change in its expense allocation policy and as a result of which the amount of expenses being charged to the unit-linked business may increase. This may reduce the statutory profits of the unit-linked business and lower the reinsurer's share of the profits and / or defer the emergence of profits to later years. In an extreme scenario, it is also possible that the delayed profit emergence may result in some of the Treaty Balance to get forfeited at the sunset date.

Therefore, the reinsurer would be interested to know more about the Company's expense policy as part of its own due diligence before executing the proposed ART solution.

[3 Marks for appropriately discussion of the reasons]

[3]

- vi) Various other ways that the Company can consider to meet its future capital requirements are discussed below:

Equity capital injection by the existing shareholders

If the Company is unlisted, capital infusion by the existing shareholders is a straightforward approach. The Company is mid-sized and its track record of profitability has been moderate. If the Company can demonstrate robust return on equity to the existing shareholders, they may be open to the idea of injecting more money into the Company. If the injection is proportional to the existing shareholding, their ownership interests in the Company would not be diluted.

Occasionally, it may be challenging to get the capital injection from the existing shareholders if the parent group does not have appropriate levels of free capital to deploy into the insurance businesses. Generally, the shareholders would inject capital in the early phases of a business and would expect the business to be self-sufficient after it becomes established in the marketplace. Since the Company is an established, mid-sized insurer, the shareholders may expect it to be self-sufficient instead of having to rely on capital injections from the existing shareholders.

There are no regulatory limits on the quantum of capital that can be raised. This is essentially subject to the financial appetite of the existing shareholders. The regulatory may in fact prefer shareholder capital injections over other sources, as this may be seen as the highest quality capital to support the business.

From a cost standpoint, equity capital is an expensive source of capital as the shareholders would typically expect a return on equity that is much in excess of the risk-free rates (e.g. with an equity spread of 4% - 5% or so).

On the other hand, unlike debt capital, there is no fixed interest servicing requirement on equity capital. The dividend yields, if any, also tend to be significantly lower in the Indian context.

Overall equity capital infusion by existing shareholders may also be perceived to be a sign of strength by the market, as it may be seen as the shareholders reposing their faith in the prospects of their business.

[1 mark for each point appropriately discussed]

[Max 5]

Raising equity capital through an Initial Public Offering ("IPO")

If the Company is unlisted, another option on the table is to carry out an IPO and raise money from the public / investors. Before considering this alternative, the Company would need to ensure that it meets the relevant regulatory criteria to launch an IPO.

The Company is already following the Indian Embedded Value approach, which is a good starting point. However, this would require to be certified by an independent actuary for the purpose of an IPO.

The Company's business plans project a capital injection in about three years' time. As such, this gives the Company sufficient time to plan out the IPO.

Market timing would be an important consideration in an IPO. Shareholders may want to launch an IPO when the general market sentiment is favorable and buoyant, as this would make it easier for the IPO to be successful. Since the Company has only a

moderate profitability track record, the shareholders may wish to strengthen the profitability before going ahead with an IPO.

Launching an IPO would be a major strategic decision requiring careful consideration. The disclosure requirements, corporate governance requirements etc. would be significantly more for a listed insurer. Managing investor expectations, both during the IPO and after the IPO, would require significant commitment from the Company's management team. The Company would wish to weigh these in carefully before launching an IPO.

The quantum of capital that can be raised may be subject to the views of existing shareholders (e.g. how much dilution of stakes are they willing to accept) and the likely valuation that the Company would attract.

From a cost standpoint, equity capital is an expensive source of capital (same point as discussed earlier). In addition, the transaction costs of executing an IPO (e.g. fees paid to bankers, lawyers, consultants etc.) may also be significant.

[1 mark for each point appropriately discussed]

[Max 5]

Preference shares or subordinated debt

Other forms of capital allowed in India include preference shares and subordinated debt. The key regulatory requirements governing these sources of capital are set out below:

- Issue of preference shares or subordinated debt is subject to a prior approval from the IRDAI;
- The instruments must be fully paid up in cash;
- The claims of the preference shareholders are superior to the claims of equity shareholders but subordinate to the claims of the policyholders and all the other creditors;
- The claims of subordinated debt holders will be superior to the equity shareholders and preference shareholders but subordinate to the claims of the policyholders and all the other creditors;
- The redemption period for the preference shares or subordinated debt should be at least ten years. Perpetual subordinated debt is permissible;
- The insurer is not liable to pay dividend or interest in any financial year if the insurer's solvency is below the minimum regulatory requirements or if the payment of dividend or interest would cause the insurer's solvency to fall below the minimum regulatory requirements;

There are regulatory limits on the amount of capital that can be raised through this source. The total quantum of capital that can be raised through preference shares or subordinated debt is capped at 25% of the equity capital (i.e. equity share capital and securities premium) of an insurer. Also, the total quantum of capital through these sources should not exceed 50% of the net worth of the insurer.

From a cost standpoint, these forms of capital are akin to debt capital. The rate of dividend or interest may be lower than the rate of return expected by equity shareholders and would depend on the credit rating of the insurer. However, the Company will need to service the capital through dividend / interest payments regularly except when its solvency falls below the minimum regulatory requirements.

Given that the servicing costs for these sources of capital are lower than the equity capital, the existing shareholders may consider this form of capital as a leverage, in order to enhance the overall return on their capital investment.

[1 mark for each point appropriately discussed]

[Max 6]

Others

- If the Company is a Joint Venture ("JV") between Indian promoter(s) and a foreign promoter and if the shareholding of the foreign promoter is less than the maximum permissible limit of 49%, the foreign shareholder may consider injecting more capital to increase its stake to 49%. This would be subject to commercial discussions between the shareholders and requisite regulatory approvals.
- If the Company is already listed, it may consider a follow-on public offer or a rights issue to augment the equity capital.
- Other capital management tools would include revisiting the new business projections (and placing limits thereon if necessary), product structure / mix (e.g. introducing products that are less capital intensive), disciplined expense management etc.
- The Company may also consider revisiting its target level of solvency ratio. If it is excessively high as compared to the regulatory requirement of 150%, it may consider lowering the target in order to lower the capital requirements.

[0.5 marks for each point appropriately made]

[Max 2]

[Max 16]

[50 Marks]

Solution 2:

- i) Key aspects of the current Indian life insurance business environment that the Global Strategy Officer should consider when preparing the preliminary 'go'/'no-go' assessment for a potential market entry focusing on protection business in India are as follows:

1. Overall regulatory environment

The Indian financial services sector (covering insurance companies, banks, non-banking financial companies, asset management companies) is governed through different regulatory bodies and government departments, as opposed to an institution that might be the case in certain other countries. The life insurance industry is regulated by the Insurance Regulatory and Development Authority of India (IRDAI) within the ambit of the Insurance Act, 1938 and Insurance Laws (Amendment) Act, 2015.

[1]

The key regulatory aspects to consider at the preliminary stage of 'go'/'no-go' assessment for market entry are set out below:

a. Market entry: registration and licensing

All life insurance companies in India must be registered with the IRDAI. New life insurance ventures are required to follow a three-stage registration procedure as set out within the IRDAI (Registration of Indian Insurance Companies) Regulations, 2000 and its subsequent amendments. The regulations require the prospective insurer to demonstrate availability of sufficient capital resources (minimum start-up capital requirements for life insurance business being INR 100 crore); adherence to foreign ownership limits (set out below); appropriateness of the promoters; establishment of fit and proper management and governance structure; provision of business plan and financial projections, amongst other things. This process could take several months to complete.

[1]

b. Foreign ownership limits

The current limit on foreign direct investment (FDI) in the insurance sector is set at 49%. Additionally, the ownership as well as control must demonstrably be “Indian” – with control being demonstrated via requirements for the Indian promoter(s) to appoint majority of the Board of Directors as well as having control over key senior management appointments and insurance company policies. This would necessitate having at least one Indian promoter to establish the joint venture (JV) life insurance company with – and consequent considerations related to selection of the appropriate JV partner and establishment of the JV etc.

[1]

c. Detailed sector-specific regulations

The life insurance industry in India is subjected to specific regulations related to most aspects of the business including: customer protection; governance and management of the insurance companies (including key senior management appointments); product pricing, design and product approvals (including requirements for minimum benefits / maximum charges / permissible features etc.); licensing and training requirements pertaining to the distribution channels and limits on commission and incentives to the distributors; marketing and advertisement; limits on expenses of management chargeable to policyholders’ funds; permissible investments and ring fencing of funds for specific lines of business; reserving and capital requirements etc.

Implications of individual regulations may be considered at a later stage. However, for the preliminary assessment, it may be worth noting, that the current regulatory environment is granular ‘rules-based’ with specific requirements for most areas of the life company operations.

[1]

2. Policyholder incentives

The Government of India offers various fiscal incentives and other benefits for the purchase of life insurance in India, in particular, related to the product lines that the multinational insurance company plans to sell, such as:

- Tax relief is available on premiums paid as well as benefits received on insurance proceeds;
- Tax incentives are provided for accumulating a ‘pension’ corpus and purchasing immediate annuities on retirement (although no tax incentives are available on the annuity income itself);
- Targeted government schemes encouraging insurance penetration.

Aside from the sector-specific incentives, broader initiatives of the Government towards financial inclusion and financial literacy; establishing a nation-wide registration of individuals and bio-metric identification; and other related schemes provide policy support to insurers to increase life insurance penetration.

3. Economic conditions

The overall macro-economic environment in India seems strong with high GDP growth rates, large and growing middle income population with high savings potential and a low level of insurance penetration. This provides huge growth opportunities to the insurer.

Currently, the interest rates are relatively high (in comparison with some other developed markets), although there is risk of interest rates falling in the future. Also, although the Government of India offers long term bonds to invest in, there is limited liquidity at several durations on the yield curve. This may need to be considered whilst designing annuity plans.

Uncertain credit market and corporate bond scenario; and limited market for derivatives / hedging instruments may also pose challenges in the management of risk-return scenario related to the annuity segment.

4. Target market / demographics and insurance demand

The high proportion of young population in India and the growing number of nuclear families (especially in urban areas) are expected to result in increasing demand for protection products.

Also, as the population ages and the number of retirees grow, it is also likely to create future demand for pensions and annuity business, especially when there is no universal social security system in India.

However, currently, there is lack of customer awareness about the need for insurance / protection and the company (along with the industry and regulator) may need to enhance the level of awareness to be able to tap the market potential.

Historically, the customer preference has also been to buy 'savings' oriented insurance plans. Although these preferences are changing gradually, the company may face challenges in the short -medium term to garner significant level of business purely focusing on 'protection' oriented products.

5. Technology

Given the possible lower average ticket size for protection business, the Company may be required to keep its overall costs low, possibly by usage of technology.

However, as compared to many other markets, India has been a hub for availability of low-cost and high quality resources to manage the operational systems and technology required to support the business. Thus, it may not be difficult for the Company to deploy innovative technological solutions in all aspects of the insurance business covering new business acquisition, underwriting, policyholder servicing, claims management as well as back-end systems such as accounting, investment and actuarial risk management.

Given that existing insurers are already competing in the protection space, the Company may need to explore opportunities to launch a predominantly 'insuretech' operation, with minimal employee strength and maximum usage of technology in order to be able to keep costs low and price products competitively.

6. Competitive Landscape

The Indian market is very price sensitive. Web aggregators – websites that compare prices across multiple insurers – are popular and driving commoditization of the

protection products in the market. Short term group insurance products are generally perceived to have thin margins due to intense competition. The margins for the group credit-linked insurance business are perceived to be healthy.

Given that the Company will be a new entrant to a market with already existing heavy competition, it may need to explore opportunities to innovate its offerings, than purely competing on pricing of the product.

However, the Company may also need to consider if it would be possible to innovate sufficiently in 'protection' space given that the products may be relatively easy to compare across insurers.

If the Company is unable to sufficiently differentiate itself on products, it may need to explore possibilities of differentiating on softer aspects such as brand recognition / value, trust factor, service standards etc. However, it may not be easy to differentiate on such factors at outset – until customers actually experience them. This aspect may need to be carefully considered by the Company.

Alternatively, the Company may need to consider strategies that focus on a niche customer segments (e.g. ultra-high net worth individuals, key-man insurance etc.)

7. Industry bodies

There is an industry association – the Life Insurance Council – which is expected to represent the interest of the life insurance companies. Once established, the Company may be able to work with the Life Insurance Council to help promote the industry through various awareness campaigns.

The Company may also be able to work with other industry bodies such as CII, FICCI to actively persuade the regulator and the Government to initiate policy reforms for future development of the industry.

Such reforms may be necessary in areas such as incentivizing the annuities (e.g. not taxing the entire annuity payment in the hands of the annuitant) or further relaxing the caps on distributor compensation applicable to the 'protection' business, in order to promote the sale of such business.

8. Distribution channels

The current market landscape includes several distribution channels including tied individual agents, bank and other third party distributors (corporate agents), brokers, online / digital as well as direct distribution channels. In the current market landscape:

- Individual protection and annuity business is sold through all channels of distribution such as tied agents, banks, brokers as well as online / digital.
- Short term group protection products are sold through insurers' direct sales teams or brokers.
- Group credit life insurance business is mainly sold through the banks and other lending institutions.
- Independent Financial Advisors (IFA) channel, which exists in some other markets, does not exist in India.

Given the need to be competitive in a price sensitive market, the Company may need to keep its distribution costs low. Hence, online / digital or direct may need to be considered as the main distribution channels for the Company.

However, the Company may also need to consider if it would be able to achieve a critical mass by focusing only on protection business through such channels? If not, the Company may be required to explore considering other distribution channels as well, including brokers.

The overall cost associated with developing a distribution channel may also need to be considered. Given intense competition in third party distribution channels (e.g. bank and other corporate agents, brokers), any such relationship may come at a high price.

9. Taxation

The life insurance industry currently enjoys a lower corporate tax rate as compared to the rates applicable to other industries. Any profits arising from the 'pensions' business is completely tax free as well. The tax losses are allowed to be carried forward and set off against profits from future years, for a maximum period of eight years.

There have been proposals to change the existing taxation regime, with proposals to increase the tax rates applicable to shareholders' share of profits from the life insurance business. However, there have not been any concrete steps taken in this respect in recent years.

There is also a tax on dividend distribution / repatriation of profits to out of India.

The insurers are also required to charge goods and services Tax (GST) on insurance premiums. However, they are allowed to offset these by any GST incurred by the insurers on its expenses.

10. Operational challenges / other considerations

The Company may need to consider the likely level of persistency experience, which may depend (amongst other things) on the distribution channel. This would be particularly important considering the 'protection' oriented strategy, with possibly high levels of initial expenses (e.g. on underwriting etc.).

Whereas the lapse rates in the industry are currently higher (than those applicable in some other markets), the Company may need to explore if it would be possible to control the same – e.g. by confining itself to online / digital / direct distribution channels to lower the risk of 'mis-selling' and through the usage of technology (e.g. through sending regular reminders for premium payments, initiating direct debits etc.)

The Indian insurers currently reinsure the pure protection business in order to benefit from the low reinsurance premium rates (thereby allowing insurers to keep the gross premiums at a low level) as well as to keep the capital requirements low. However, there may be a risk of the reinsurance rates hardening in the future. Unless the Company is able to re-price the products, the overall profitability of the business may come under pressure. The Company may need consider this risk whilst exploring the market entry opportunity.

[2.5 marks for each of the points from (2) to (10) above, appropriately explained]

[Max 14]

[Max 18]

ii) The following corrections / refinements are potentially needed to the proposed approach, to fully comply with the requirements of the regulations and the applicable professional standards (APS7):

1. Lapse/surrender rates are increased by 20% for a margin over current best estimates. This assumes that an increase in lapse / surrender rates is more prudent and would lead to an increase in reserves. The Company would need to test whether this indeed results into an increase in reserves and if not, allow for a reduction in the assumed lapse / surrender rates below the current best estimates. Also, instead of testing this only at the valuation date, the Company may wish to test this at future policy durations as well, to ensure that the assumed lapse / surrender rates in future policy years also allow for appropriate level of prudence; [1.5]
2. Currently no margin for prudence seems to have been applied to future expenses. The Company may need to reflect a margin of at least 10% over the best estimate expense levels;
3. Also, the assumed per policy expense loadings are set based on their long term steady state level. However, the regulations require that the reserves should allow for the actual expenses of the Company. Where the actual expenses are not used, the regulations require an additional provision be held in respect of any possible shortfall due to the implied maintenance expense over-run (subject to a carve-out that this is not required in the first five years of the commencement of the business). Hence, the Company may need to ascertain whether the assumed long term steady state expense loadings are expected to be achieved within five years or not. If not, then an additional reserves may be required to be held after five years to allow for the level of maintenance expense overruns that may be applicable then;
4. The expense inflation assumption should be internally consistent with the assumed valuation interest rate. At present, both the expense inflation assumption and the valuation interest rate are set at 6.0% p.a. implying a nil real rate of interest. Although plausible, a sustained nil real rate of interest may be against conventional economic wisdom, and therefore this assumption may be revisited; [4]
5. The valuation rate of interest is required to be set by having regard to the yield on existing assets, allowing for a future reinvestments and then reflecting a margin for adverse deviations. However, since there are no existing assets at present, an approach using an estimate of ultimate risk free forward rate is not unreasonable. However, the Company may consider setting the 'best estimate' investment returns expected to be earned on the assets backing the future liabilities and then reflect a margin for adverse deviation in line with the APS7 requirements;
6. In addition to allowing for (re)investment risk and a margin for adverse deviation, the valuation rate of interest must also allow for any expenses associated with the investment function of the insurer. Therefore, an allowance (whether implicit or explicit) for investment expenses may need to be incorporated in the valuation interest rate; [2]

7. The APS7 only specifies 'minimum' adverse scenarios whilst setting the margins for prudence in the reserving bases. It also suggests that if there is lack of credible experience to set the 'best estimate' assumptions, a higher level of margin for prudence may be considered whilst setting the reserving bases. Given that the Company is yet to start its operations and therefore lacks any experience, the Company may consider adopting a higher level of margins than those adopted. [1.5]
 8. The regulations / actuarial practice standards also require additional 'global reserves' to be held to allow for potential strains arising due to items such as incurred but not reported (IBNR) claims, expenses in a new business discontinuance scenario, costs of any options and guarantees, any potential revival strains for currently lapsed policies etc. Given that these are not considered in the reserving approach adopted, the Company may need to allow for them separately in order to be able to estimate the future capital requirements appropriately. [1.5]
 9. The reserves are floored to zero at a policy level, in case of negative net present value. A further surrender value floor would also need to be introduced, where policies have acquired a surrender value (which may be the case, particularly for the term assurance with a return of premium at maturity) [1]
 10. The solvency capital is computed at the current applicable factors. However, there is also an absolute amount of on-going solvency requirement (of INR 40 Crores). Also, it is not clear whether the regulatory requirement to hold a minimum of 150% of the required solvency margin is considered in the current analysis. Given that the objective is to estimate the overall capital requirements for the Company, appropriate adjustments may be required to the calculated solvency capital. [2]
- iii) The various sources of profits in term insurance contracts and the possible reasons for a) lower term assurance premium rates in India than other Asian / global markets despite similar population mortality rates; and b) relatively higher value of new business (VNB) margins in protection than other lines of business are discussed below [Max 12]

1. Mortality

- a. There is a significant heterogeneity amongst the Indian population – with large differences in the education levels, income levels, access to healthcare etc. Naturally, this results in significant differentiation in the mortality levels across various strata of the society. It is possible that as compared to other Asian / global markets, the Indian insurers may be targeting a different / select profile of customer segment and not that represented in the mortality experience of the general population. Typically, the people on the lower end of the socio-economic strata are either uninsured or insured through Government schemes, employer-employee group schemes etc., whereas the insurers may be targeting middle class and high net worth segments. As such, the insured customers' mortality experience may not be appropriately reflective of the general population mortality.

This may be further accentuated if a majority of the individual term assurance business sold in India ends up being medically underwritten. This may allow the insurers to appropriately segment risks by charging standard rates to healthy lives and loading underwriting extra premiums for less healthy lives. On the other hand, the term assurances sold in other Asian / global markets may be processed as

“straight through”, using simplified underwriting, leading to the mortality experience being closer to the population mortality rates.

- b. Assuming that both the pricing and VNB margins calculation reflect the expected mortality rates for all LOBs, one would expect the mortality assumption to be the more dominant factor in determining the overall profitability (and therefore the VNB margin) of term insurance business as compared to other LOBs. The average policy size of a term insurance plan is typically lower than that for other LOBs. Thus, even if the actual mortality profits are at similar level across all LOBs, the term insurance VNB (which is expressed as a % of premiums charged) are higher than other LOBs.
2. Expenses
 - a. The overall expenses could be a cause of the “low” premium rates in India as compared to Asian / global levels, if the expense / commission levels in India are materially lower than these markets. Indeed, there are caps on the overall commission levels in India which may be lower than those applicable in other markets. Furthermore, the term assurance products sold online / digitally are “direct-to-customer” and therefore do not have any loadings for commissions. Admittedly, there may be IT costs incurred in setting up the online / digital distribution channel. However, these IT costs may be much lower in India as compared to other Asian / global markets.
 - b. Several insurers in India are still experiencing expense overruns, wherein the actual expenses are much higher than those allowed for in pricing. Assuming that the VNB calculations reflect the ‘actual’ experience, this is typically a source of ‘loss’ across all LOBs. The extent of such loss in each LOB would depend on the actual expenses allocated to each LOBs. If the allocated expenses to other LOBs are higher than those allocated to term insurance business, the term insurance business may look relatively profitable than the other LOBs.
3. Lapses / surrenders
 - a. The actual lapse / surrender rates in India may be higher as compared to some other Asian / global markets. The term assurance products in India are typically lapse-supported – as the regulations mandate level premium structure and insurers typically do not offer surrender values. If the pricing of term insurance products in India assume a high level of lapses rates, this could be a potential reason for the low premium rates relative to other Asian / global markets.
 - b. When compared to term assurances, the extent of lapse profitability of other LOBs is quite different. Unit-linked products generally result into losses upon lapses / surrenders, due to the restrictions on the surrender charges. In participating savings products, any lapse profits remains into the participating fund, limiting the shareholders’ share of such profits to 10%. Depending on how the products are priced, the non-participating savings products may also have lower reliance on lapse profits than the term assurances.
4. Investment returns
 - a. The level of interest rates in India may be at a higher level than those in Asian / global markets. Thus, the pricing of term insurance contracts, performed assuming a higher investment return, would result in a lower premium rates as compared to other Asian / global markets. Although less dominant a factor for shorter term

insurance (which do not result into much reserve build up), this may be a more dominant factor for longer term insurance contracts.

- b. The contribution from investment profits to the overall VNB of term insurance contracts may not be high in comparison to that from other LOBs. However, for longer term assurance contracts (with sizeable reserves build up), this may be a non-trivial source of contribution to VNB, helping the overall VNB for term products to be higher than those for other LOBs.
5. Reinsurance
 - a. The reinsurance terms could be a significant source of difference across countries. In India, the low reinsurance premium rates charged help the insurance companies keep their own premium rates at a low level. Reinsurance terms are essentially a source of additional profits for term insurance contracts in India. Reinsurance also significantly reduces the level of required capital, thereby lowering the cost of capital (if the VNB margins are measured following a traditional approach). This may not be the case in other Asian / global markets or may not be the case to the same extent as in India.
 - b. When compared to term assurances, the reinsurance benefits may be limited for other LOBs. Given the relatively high retention limits generally adopted, it is likely that the mortality risks within a vast majority of the new business though savings-oriented LOBs may be fully retained by the insurers.
6. Allowance / loading for risk / loading for cost of capital
 - a. Any allowance / loading for risks / loadings for cost of capital as reflected in the premium rates charged to the policyholders may be lower in India as compared to other Asian / global markets. This may be due to the fact that a large proportion of term insurance business is reinsured in India and hence the insurer may not reflect a high loading for risks retained.
 - b. The manner in which an allowance / loading for risk / loadings for cost of capital is reflected in pricing (e.g. through the usage of risk discount rate) may be different as compared to that reflected in the calculation of VNB (typically only frictional cost of capital, 'FC'; and an allowance for cost of residual non-hedgeable risks, 'CRNHR' in case of market consistent valuation). This difference in allowance for risk would unwind into VNB margin calculation. The impact of this may be more muted for other LoBs than for term insurance contracts.
7. Explicit allowance for profit loading / profit criteria
 - a. It is possible that the large untapped opportunity for insurance business in India, which has attracted many insurers, has also resulted in a high level of competition and driven down the profit loadings to inherently lower levels relative to other Asian / global markets. Particularly, profit loadings in global markets may be higher as the markets may have matured towards oligopolistic states, leading to lower competitive intensity.
 - b. The profitability profiles across different LoBs are impacted by different factors – for unit-linked, they are restricted due to cap on charges; for participating, they are restricted due to 90/10 profit sharing gate; and for non-par savings, they are restricted due to available spread between expected return on investments v/s return required to be given to the policyholder due to competitive pressure. Thus,

insurers may be reflecting a higher profit loading in term insurance contracts as compared to other LOBs where the profits loadings are implicitly / explicitly restricted by regulation or competitive pressures.

[2 marks for each of the 7 points listed above appropriately discussed]

[Max 10]

iv) The arguments in favour of acquiring stakes in an existing life insurance company are:

1. **Limited / non-availability of suitable joint venture partners**

- Considering the protection products strategy that the MNC wishes to pursue, the new insurer to be set up may need to keep its distribution costs low and thus need to have access to an existing distribution network, instead of setting up expensive distribution channels (e.g. individual agency). Thus, the MNC may be looking for a tie-up with a domestic promoter that provides such a distribution network to the new insurer.
- At the same time, the domestic promoter needs to meet all the 'fit and proper' criteria (including ability to support capital requirements of the new insurer) specified by the IRDAI for the new insurer to be able to get a license from the regulator.
- It may be very difficult for the MNC to find such a partner in India considering that the suitable / willing partners may have already set up their own life insurance companies. As such, the MNC may consider more appropriate to take a stake in an existing life insurance company than trying to find a suitable joint venture partner.

[2]

2. **Competition**

Given the focus on protection / annuity business, the new insurers may also be required to be empaneled with the Pension Fund Regulatory and Development Authority (PFRDA) to provide annuities in case of NPS and invest significantly in brand building activities. Given the existing level of competition in both the protection and annuity space in India, this may act as an entry-barrier for a potential new entrant.

[1]

3. **Speed to market**

Acquiring stakes in an existing life insurance company may be faster than going through the process of finding a domestic joint venture partner, negotiating the joint venture terms, going through the IRDAI's licensing process and establishing and operationalizing the new insurance venture.

4. **Operational / distribution synergies**

- An existing life insurer would already have a set up and active operation. The MNC would not be required to "re-invent the wheel" and invest heavily in initial years in setting up the operation.
- An existing life insurer might even get some good distribution capabilities based on its existing relationships, which may either not be possible or may be time consuming / costly for a new entrant.

[1]

On the other hand, the arguments against acquiring stakes in an existing life insurance company and in favour of setting up a 'greenfield' operation are:

1. Speed to market may not be short

Acquiring stakes in an existing life insurance company may be a lengthy and cumbersome process. For example:

- The MNC may first need to identify an insurer whose strategic directions are aligned to that of the MNC (i.e. who wishes to focus only on protection and annuities) and carry out appropriate due diligence on the target insurer;
- Considering the FDI cap in insurance sector, the foreign promoter in the target life insurance joint venture, if holding the maximum 49% permitted, may first need to be willing to exit for the MNC to be able to take stakes in the target insurer;
- In case the domestic promoter(s) of an existing insurance company also wishes to exit, this may complicate the process further, as the MNC may then need to find a replacement domestic promoter;
- The negotiation process to agree on suitable terms for acquiring stakes may not be straight forward. There may be a mismatch between the price expectation from the 'seller' and the fair value estimation by the MNC;
- The regulatory approval process itself may not be seamless.

[2]

2. Avoidance of legacy issue risks

- Despite the due diligence that may be performed before taking a stake in an existing life insurance company, there may still be risks (e.g. compliance issues / operational issues / accumulated fraud risks etc.) inherent in the business which may hit the insurer several years after the acquisition of stakes by the MNC. Depending on the nature of such issues, it may risk the insurer incurring significant additional costs (e.g. compensation for mis-selling), damage to the reputation and diversion from the strategic focus.
- For a new venture, the MNC would be able to shape the culture, processes and practices of the insurer from inception itself and hence it may be easier to control any malpractices and avoid such risks.

[1]

3. Focused strategy

- The MNC is aiming to adopt its global strategy of focusing on protection business in India. Taking stakes in an existing insurance company (which may have been selling other types of products such as unit-linked, participating, non-participating savings) may not allow the MNC to implement this strategy immediately and in totality.
- The MNC may need to either divest these lines of businesses or run them off as a closed book or continue writing these product lines, all of which would result in either additional costs or dilution of its focus. The MNC may also be required incur additional costs in 're-branding' / communicating the revised focused areas for the insurer.
- Any inefficiencies in the existing distribution / operational aspects may also need to be addressed as any cost inefficiencies arising due to these may hamper the intended 'protection' business focus of the MNC.
- In comparison, rolling out the global strategy in a new venture may be much more seamless and cost effective.

[2]

[Max 2 marks for any other appropriate points discussed]

[Max 10]

[50 Marks]
